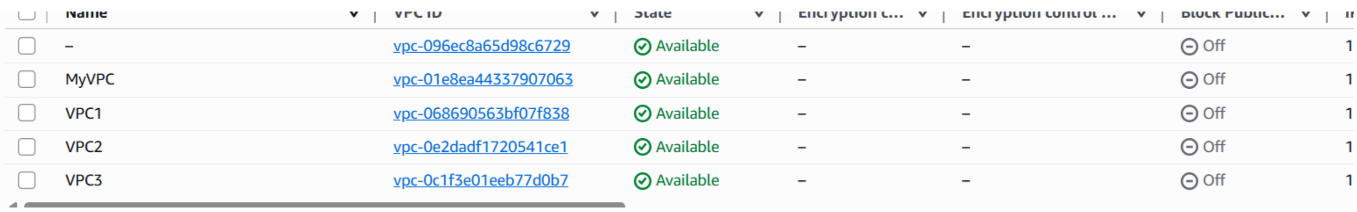
VPC Challenge

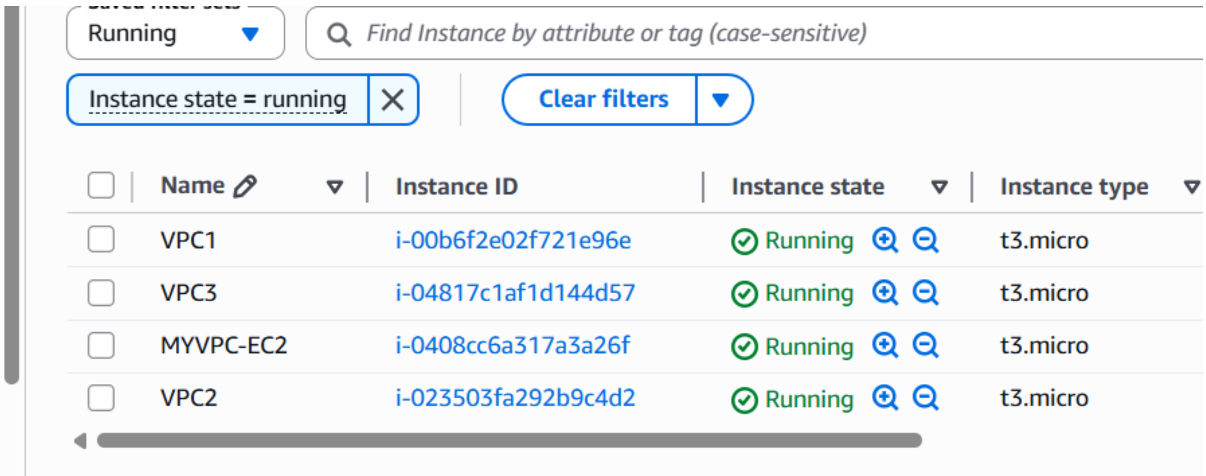
Use Case: Setting up Transit Gateway and VPC Endpoints for a Multi-VPC Architecture
Scenario:
A large organization is migrating its on-premises infrastructure to the AWS cloud.
The organization's architecture involves multiple VPCs for different departments and applications, each requiring secure communication with centralized services and external resources.
The IT team needs to design and implement a scalable and efficient network architecture to accommodate the organization's growth and ensure robust connectivity between VPCs and external services.
Objectives:

1. Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.
Created 4 vpc's with names MYVPC/VPC1/VPC2/VPC3, and added Internet gateway, created subnets for each vpc and added to the Route Table,
Created EC2 instance for each vpc and added security groups also to it



Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	I
–	vpc-096ec8a65d98c6729	Available	–	–	Off	1
MyVPC	vpc-01e8ea44337907063	Available	–	–	Off	1
VPC1	vpc-068690563bf07f838	Available	–	–	Off	1
VPC2	vpc-0e2dadf1720541ce1	Available	–	–	Off	1
VPC3	vpc-0c1f3e01eeb77d0b7	Available	–	–	Off	1

4 VPC public instances created



Running

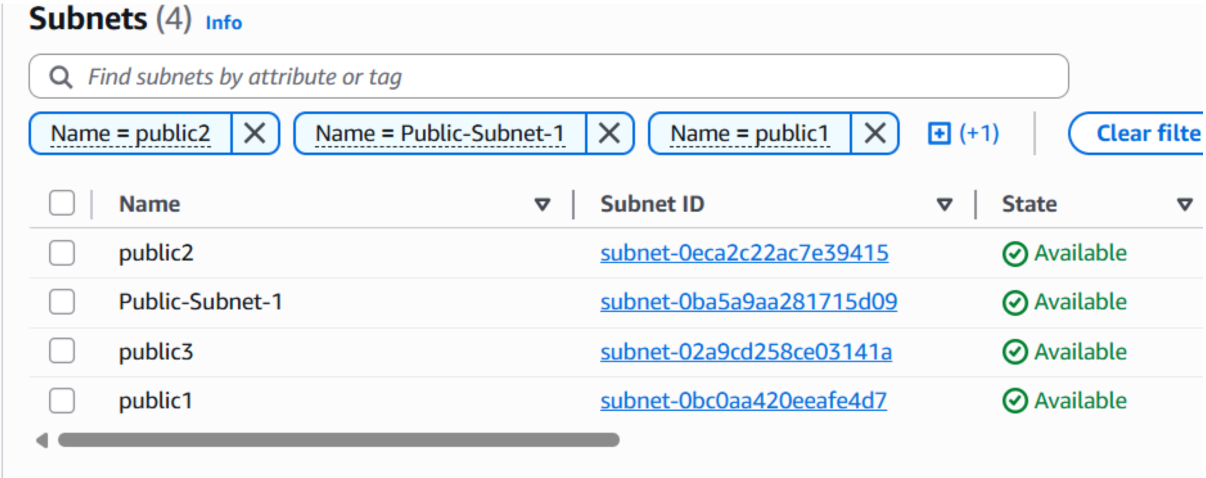
Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

Name	Instance ID	Instance state	Instance type
VPC1	i-00b6f2e02f721e96e	Running	t3.micro
VPC3	i-04817c1af1d144d57	Running	t3.micro
MYVPC-EC2	i-0408cc6a317a3a26f	Running	t3.micro
VPC2	i-023503fa292b9c4d2	Running	t3.micro

Public subnets created



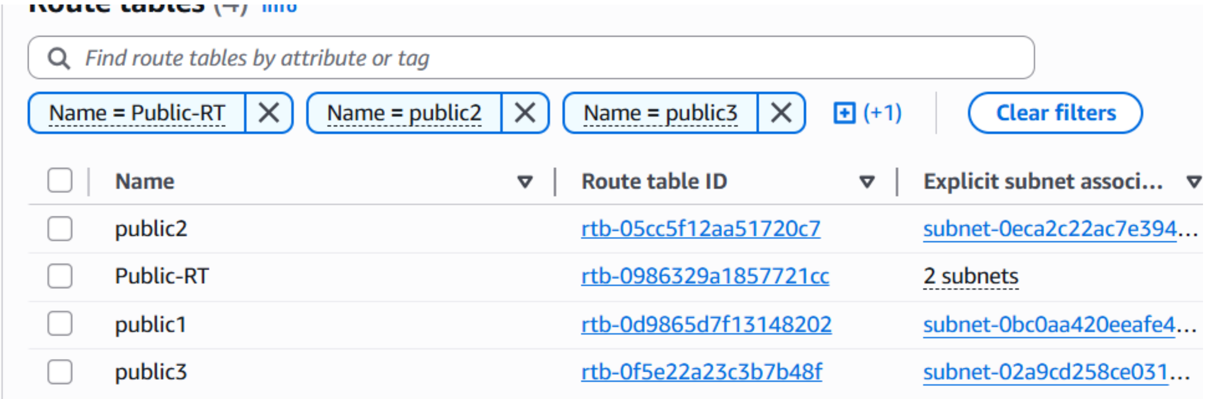
Subnets (4) Info

Find subnets by attribute or tag

Name = public2 | Name = Public-Subnet-1 | Name = public1 | (+1) | Clear filte

Name	Subnet ID	State
public2	subnet-0eca2c22ac7e39415	Available
Public-Subnet-1	subnet-0ba5a9aa281715d09	Available
public3	subnet-02a9cd258ce03141a	Available
public1	subnet-0bc0aa420eeafe4d7	Available

In Route Tables 4 public routes created

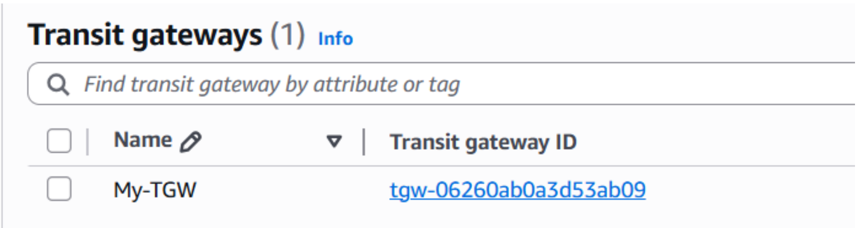


Find route tables by attribute or tag

Name = Public-RT | Name = public2 | Name = public3 | (+1) | Clear filters

Name	Route table ID	Explicit subnet associ...
public2	rtb-05cc5f12aa51720c7	subnet-0eca2c22ac7e394...
Public-RT	rtb-0986329a1857721cc	2 subnets
public1	rtb-0d9865d7f13148202	subnet-0bc0aa420eeafe4...
public3	rtb-0f5e22a23c3b7b48f	subnet-02a9cd258ce031...

Created transit gateway

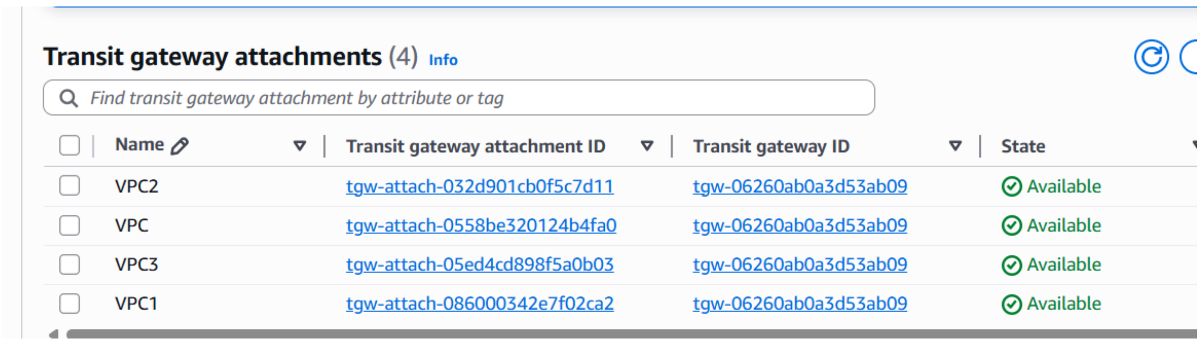


Transit gateways (1) Info

Find transit gateway by attribute or tag

Name	Transit gateway ID
My-TGW	tgw-06260ab0a3d53ab09

And here created a transit gateway attachment to vpc's



Transit gateway attachments (4) Info

Find transit gateway attachment by attribute or tag

Name	Transit gateway attachment ID	Transit gateway ID	State
VPC2	tgw-attach-032d901cb0f5c7d11	tgw-06260ab0a3d53ab09	Available
VPC	tgw-attach-0558be320124b4fa0	tgw-06260ab0a3d53ab09	Available
VPC3	tgw-attach-05ed4cd898f5a0b03	tgw-06260ab0a3d53ab09	Available
VPC1	tgw-attach-086000342e7f02ca2	tgw-06260ab0a3d53ab09	Available

Attached the created transit gate ways to the route tables

	local	
172.168.0.0/24	Peering Connection pcx-05623a853bbe4ba8a	Active
132.1.0.0/16	Transit Gateway tgw-06260ab0a3d53ab09	Active
152.2.0.0/16	Transit Gateway tgw-06260ab0a3d53ab09	Active
122.1.0.0/16	Transit Gateway tgw-06260ab0a3d53ab09	Active
0.0.0.0/0	Internet Gateway igw-0604070e37ed1a657	Active

Ping the <private ip> of the other instances and it's working properly. So all the VPC configs and Transit gateway has been configured properly.

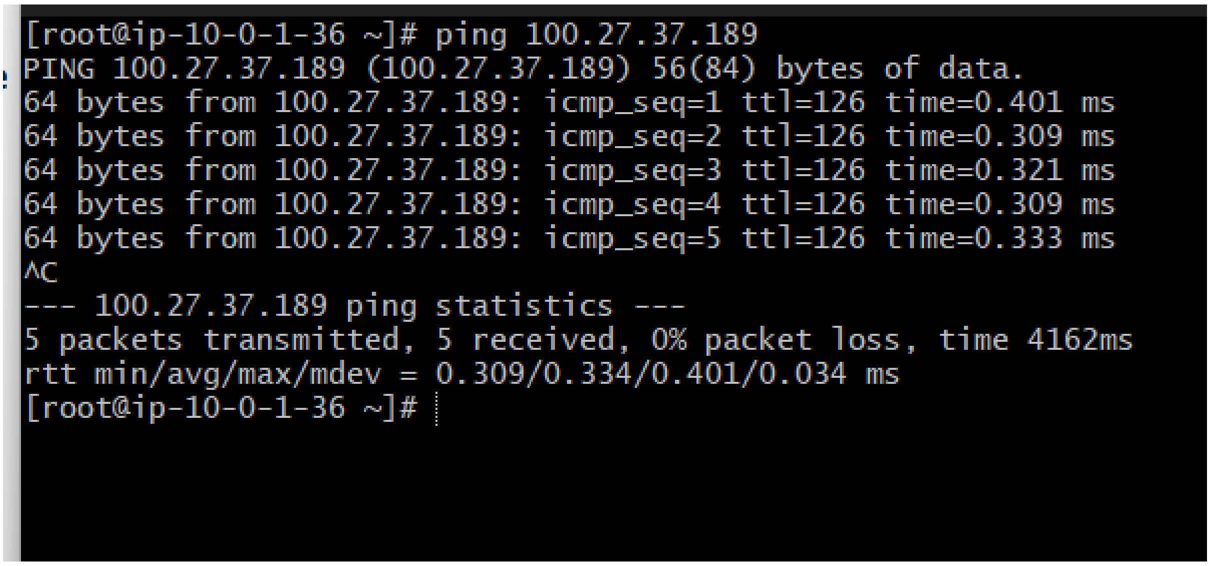
MYVPC to VPC1 Public

```
[root@ip-10-0-1-36 ~]# ping 44.192.59.183
PING 44.192.59.183 (44.192.59.183) 56(84) bytes of data.
64 bytes from 44.192.59.183: icmp_seq=30 ttl=126 time=0.339 ms
64 bytes from 44.192.59.183: icmp_seq=31 ttl=126 time=0.358 ms
64 bytes from 44.192.59.183: icmp_seq=32 ttl=126 time=0.417 ms
64 bytes from 44.192.59.183: icmp_seq=33 ttl=126 time=0.310 ms
64 bytes from 44.192.59.183: icmp_seq=34 ttl=126 time=0.305 ms
64 bytes from 44.192.59.183: icmp_seq=35 ttl=126 time=0.303 ms
64 bytes from 44.192.59.183: icmp_seq=36 ttl=126 time=0.305 ms
64 bytes from 44.192.59.183: icmp_seq=37 ttl=126 time=0.327 ms
^C
--- 44.192.59.183 ping statistics ---
37 packets transmitted, 8 received, 78.3784% packet loss, time 37441ms
rtt min/avg/max/mdev = 0.303/0.333/0.417/0.036 ms
[root@ip-10-0-1-36 ~]#
```

MYVPC TO VPC2 Public

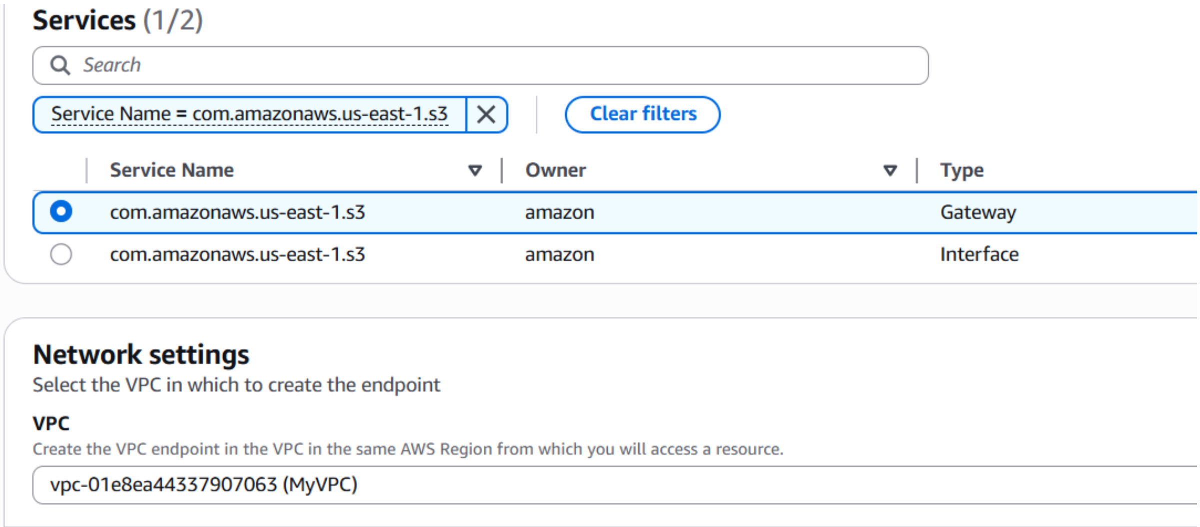
```
root@ip-10-0-1-36:~
[root@ip-10-0-1-36 ~]# ping 3.221.150.169
PING 3.221.150.169 (3.221.150.169) 56(84) bytes of data.
64 bytes from 3.221.150.169: icmp_seq=46 ttl=126 time=0.310 ms
64 bytes from 3.221.150.169: icmp_seq=47 ttl=126 time=0.319 ms
64 bytes from 3.221.150.169: icmp_seq=48 ttl=126 time=0.291 ms
64 bytes from 3.221.150.169: icmp_seq=49 ttl=126 time=0.284 ms
64 bytes from 3.221.150.169: icmp_seq=50 ttl=126 time=0.313 ms
64 bytes from 3.221.150.169: icmp_seq=51 ttl=126 time=0.299 ms
```

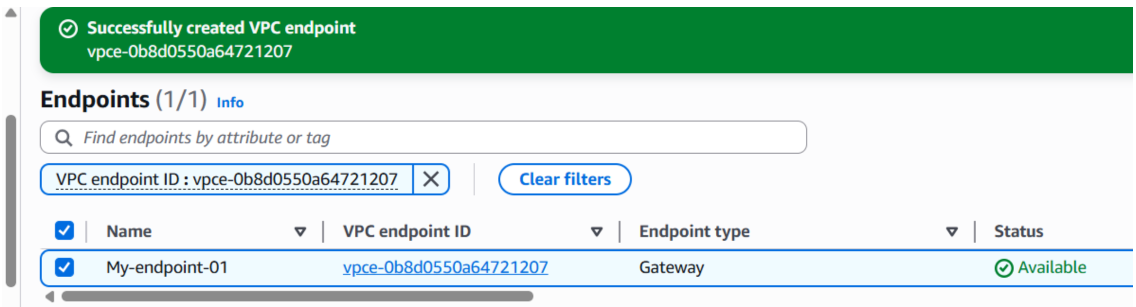
MYVPC TO VPC3 Public



2. Configure VPC endpoints to securely access AWS services without internet gateways or NAT gateways, ensuring data privacy and minimizing exposure to external threats.

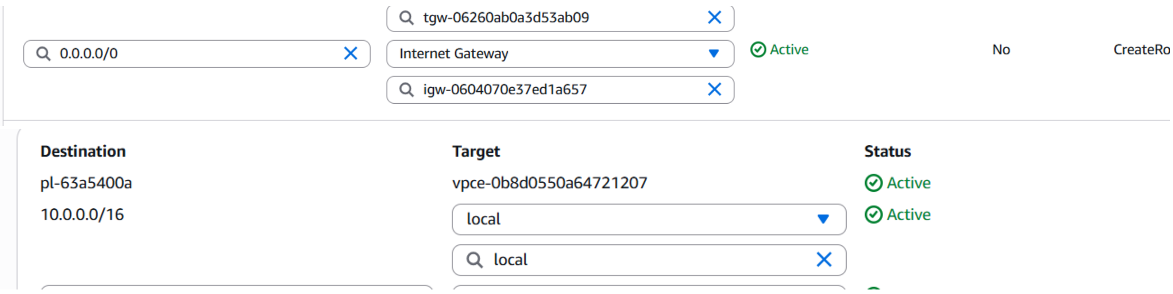
Created a Gateway VPC Endpoint for Amazon S3





Configured Route Table for Endpoint

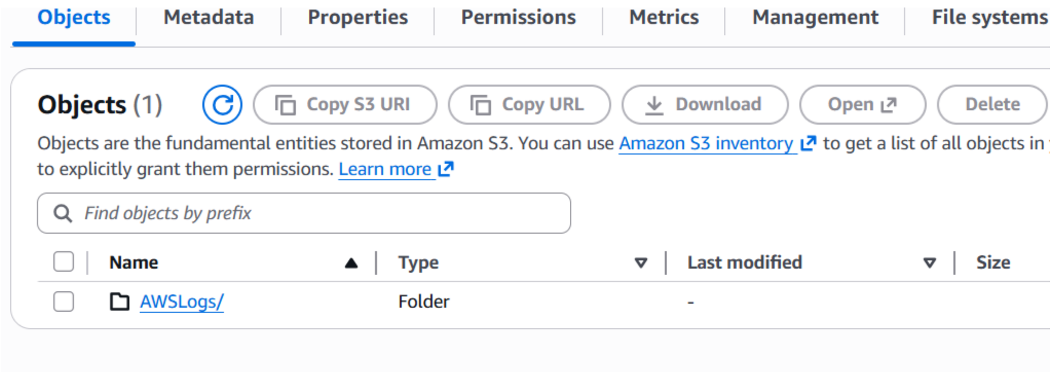
- **The route table was automatically updated with a prefix list (pl-xxxx).**
- **Destination traffic to S3 is routed via the VPC Endpoint instead of the internet.**



Enabled Private Access Without Internet/NAT

- **EC2 instances accessed S3 without using Internet Gateway or NAT Gateway.**
- **No public IP address is required for accessing S3.**

Created S3 bucket



Verified Connectivity Using AWS CLI

- **Configured AWS CLI on EC2 instance.**
- **Executed command:**

```
[root@ip-10-0-1-36 ~]# aws s3 ls

Unable to locate credentials. You can configure credentials by running "aws login".
[root@ip-10-0-1-36 ~]# aws configure
AWS Access Key ID [None]: AKIA5QKV7AB3RRBO2LN3
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
[root@ip-10-0-1-36 ~]# aws s3 ls
2026-04-15 15:15:35 my-vpc-flowlogs-bucket-123
2026-04-15 16:02:26 my-vpc-flowlogs-bucket-456
[root@ip-10-0-1-36 ~]#
```

- **Successfully retrieved list of S3 buckets.**
- **This confirms endpoint connectivity is working properly.**